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Motion and Representation Influence Numeriacal-Experimental Agreement in Fish Wake Pressure

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Swimming fish generate hydrodynamic signatures that may be detected by other animals, but it remains unclear which details of body motion must be represented to reproduce those signatures. To test this, pressure measurements from three wall-mounted sensors were combined with two-dimensional numerical wake models driven by observed trajectories of striped eel catfish, *Plotosus lineatus*. A deforming fish-shaped digital twin was compared with a cylinder following the same measured centre-of-mass paths. Measured pressure segments were more consistent with the deforming fish model than with the cylinder, especially when boundary motion along both tank axes was communicated to the flow solver. With 95% of simulated pressure variance retained and a 0.99 classification-confidence threshold, this model classified 30.3% of measured segments as fish-like and 4.4% as cylinder-like. Distributional analysis of pressure-conditioned vortices showed that cylinder-associated vortices were larger, had higher mean core Q and occupied a greater fraction of their shorter pressure segments, while fish-associated vortices were more numerous and followed straighter trajectories. These results show that retaining fine-scale body deformation improves preservation of pressure signatures measured from swimming fish.

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1. Introduction

Swimming animals generate hydrodynamic wakes composed of vortices, velocity fluctuations and pressure variations. The structure of these wakes can reflect properties of the generating motion, body size, trajectory and recent movement history [1]. Fish can detect local hydrodynamic disturbances through the lateral-line system, which responds to pressure gradients and water motion associated with nearby flow structures [2, 3, 4]. Fish can also exploit vortices in collective and wake-following contexts [5]. Hydrodynamic wakes are therefore potentially information-rich sensory cues, although the contribution of different components of body motion to those cues remains difficult to isolate in freely swimming animals.

A related body of work treats wake sensing as an inverse problem. Rather than predicting the flow generated by a known body, properties of the source are inferred from hydrodynamic measurements. Neural networks have classified canonical wakes behind oscillating foils from local vorticity measurements [6], and artificial neural models have identified idealised obstacle shapes from distant pressure or velocity fields in potential flow [7]. Artificial lateral-line systems have also combined body-surface pressure with neural networks to distinguish the shape and dimensions of moving underwater targets in simulations and experiments [8]. These studies show that local measurements can contain recoverable source-related information, but they rely on canonical, idealised or prescribed target flows. Whether sparse pressure measurements can distinguish signals associated with a freely swimming biological source from those of a simplified physical surrogate has received less attention.

Experimental access to the full wake is usually limited because flow measurements around living animals can perturb the animal or cover only part of the domain. A common strategy is therefore to replace the swimmer with a simplified body, such as a cylinder, flexible foil, finite-dipole swimmer or few-link articulated swimmer. These surrogates isolate elementary wake mechanisms and make simulation or experimental control tractable, but they simplify different aspects of the animal. In collective-motion and navigation models, finite-dipole models deliberately abstract both morphology and kinematics [9], while reinforcement-learning studies may reduce the body to a three-link fish in potential flow, with deformation represented by two hinge angles [10]. Higher-fidelity reinforcement-learning simulations can resolve vortical interactions around self-propelled swimmers, but still focus mainly on energetic optimization in controlled leader-follower steady swimming settings [11]. Recent robotic-fish studies have added richer control and sensing, including LSTM-based navigation through a cylinder-generated Karman vortex street [12] and pressure-informed, transformer-based control of a serially connected rigid swimmer in a free stream [13]. These approaches increase hydrodynamic and control realism, but their body motion and deformation remain more constrained than measured living-fish deformation.

More anatomically detailed simulations reduce geometric simplification by including realistic fish form and kinematics. Comparisons that exchanged carangiform and anguilliform forms and kinematics among self-propelled virtual swimmers showed distinct effects of form and motion on speed, efficiency and small-scale wake structure [14]. More recent wall-resolved simulations of lamprey and mackerel found swimmer-specific pressure and vorticity scaling, illustrating how body form and swimming mode jointly shape the surrounding flow [15]. Jack-fish simulations showed that real kinematics can change hydrodynamic performance relative to prescribed motions, while still simplifying the trajectory to steady swimming and controlled wavelength variation [16]. Flow-physics-informed models of carangiform schools use wake information from high-fidelity simulations to predict beneficial relative positions, but likewise assume steady swimming and prescribed kinematic or positional parameters [17]. Fish-like NACA profiles and robotic platforms provide strong control over gait, tailbeat phase or body-wave prescription [18, 19]. Across these approaches, increasing geometric detail does not necessarily preserve the unconstrained motion of a living swimmer.

Together, these literatures leave two connected questions. The first is whether sparse pressure signals contain enough source-related information to separate a deforming fish-shaped model from a simple physical surrogate. Establishing that separation is necessary before the relative importance of trajectory, body profile and deformation can be evaluated. The second question is which parts of a living fish's measured motion must be represented to preserve pressure and vortex signatures. A rigid surrogate may reproduce the centre-of-mass path or broad wake periodicity while omitting local surface motion and projected shape changes. Conversely, additional digital-twin detail is useful only if it improves agreement with measurements.

Here, a deforming fish-shaped digital twin and a cylinder driven by identical experimentally observed centre-of-mass paths were compared. Pressure measurements from three wall-mounted sensors provided a sparse, non-invasive reference. This made it possible to test whether measured pressure segments were more consistent with the fish-shaped model than with the trajectory-matched cylinder, and whether contour-size regularisation and inclusion of boundary motion along both tank axes changed that consistency. A classifier trained only on simulated fish and cylinder pressure segments provided a comparative metric. Throughout, "fish-like" means more consistent with the deforming digital twin in the learned pressure-signal space, but it does not imply direct reconstruction of the experimental wake or behavioural recognition by another fish.

2. Materials and methods

(a) Experimental data and sample accounting

The data originated from infrared video recordings of 40 striped eel catfish, *Plotosus lineatus*, each recorded in one video while swimming in a water tank equipped with three wall-mounted pressure sensors. Recordings were acquired at 100 frames s^{-1} with an image resolution of 2000 $\times$ 1200 pixels. Fish body contours and sensor locations were visible in the same view, allowing fish motion to be related to pressure measured at the wall.

The 40 videos were segmented into 493 discrete samples in which a fish was visible. Fish contours were extracted with the HAACHAMA pipeline [20, 21], summarised in Figure 1. Of the 493 segmented samples, 403 (81.7%) were retained: 367 samples processed without manual intervention, 17 requiring heavy temporal cropping, six with some poor segmentation and 13 containing a temporary circular-segmentation error. The remaining 90 samples were not retained. This filtering was completed before simulation and classifier analyses. Full sample accounting and exclusion categories are reported in supplementary table S1.

(b) Digital twins and simulation variants

Each retained video frame was represented by a closed fish contour, and each sequence described both displacement and body deformation through time. The contours generated a deforming fish-shaped digital twin in LilyPad [22]. The same measured centre-of-mass trajectory drove a cylindrical surrogate whose diameter was set from fish body width. The two bodies therefore followed the same observed path but differed in geometry and deformation.

The simulation coordinate system was fixed to the tank: x was the long tank axis and y was the transverse tank axis. Because fish swam predominantly along x, x motion was approximately streamwise and y motion captured most lateral undulation, but these were laboratory-coordinate components rather than instantaneous fish-centred axes.

Two uncertainties motivated the simulation variants. First, the independently segmented contours fluctuated in projected size between frames. These changes could represent segmentation noise, natural posture variation or out-of-plane motion such as body roll. A fixed profile based on the first frame was therefore tested to determine whether regularising projected size improved agreement with measured pressure. Conceptually, this comparison examined how small changes in shape affected the wake. Second, the original implementation treated swimming as progression mainly along x with deformation mainly along y, following a common approximation for idealised undulatory swimming. Real trajectories were not perfectly aligned with x, and local body motion could also contain an x component when the fish turned or when the tail tip moved towards or away from the head. Two-component boundary forcing was tested to determine whether these secondary in-plane motions contributed to the pressure signature. From a high level, this comparison examined how fully representing the deformation of the shape affected the wake. The two tests produced four variants, illustrated in Figure 2:

(i) *Raw*: the original image-derived contours with the original y-only boundary forcing.
(ii) *Size-constrained*: a profile based on the first frame was fitted along the reconstructed spine in every frame. This preserved bending while constraining projected size, and the original y-only forcing was retained.
(iii) *X-deformation*: the original contours were retained, but both x and y boundary-motion components forced the fluid.
(iv) *Full*: the size constraint and two-component boundary forcing were combined.

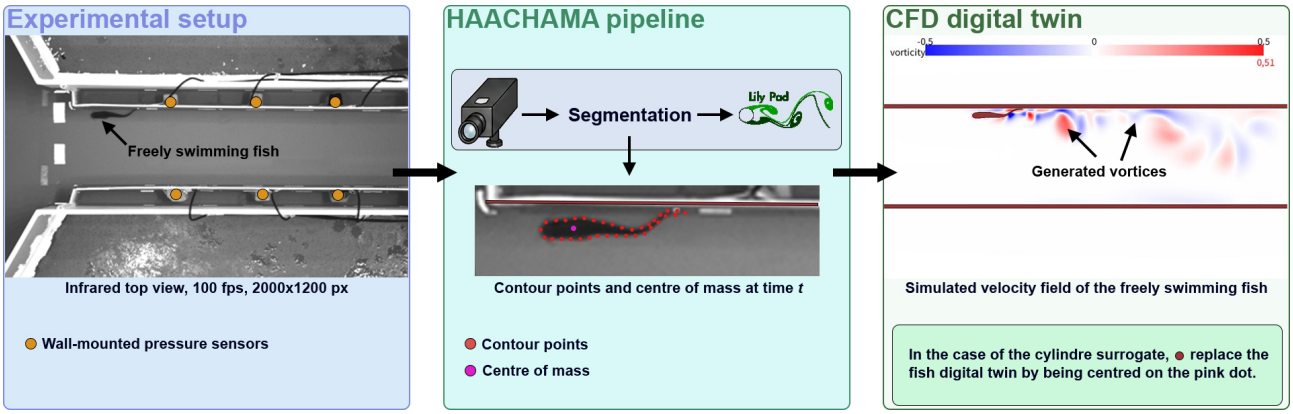


Figure 1. Overview of the HAACHAMA pipeline used to generate a deforming fish-shaped digital twin and a cylinder surrogate. Infrared video provided the fish trajectory and body contour, while three wall-mounted sensors recorded experimental pressure. The fish-shaped digital twin and cylinder were driven by the same centre-of-mass trajectory. The pink centre-of-mass marker is a conceptual annotation added to help relate the experimental and simulated representations. It is not an output generated by the flow simulation or a direct result of the segmentation.

For the fixed cylinder geometry, the size constraint had no effect. Consequently, the raw and size-constrained cylinder simulations were identical, as were the x-deformation and full cylinder simulations. Further implementation details are provided in supplementary sections S6 and S7.

(c) Simulated vortices and pressure segments

The experiment provided pressure at three wall locations rather than a spatially resolved wake field. Corresponding simulated pressure values were extracted at those three sensor coordinates during the passage of tracked vortices. Vortices were identified from simulated velocity fields using the Q-criterion [23, 24],

$$Q = \frac{1}{2}\left(\|\Omega\|^2 - \|S\|^2\right), \tag{2.1}$$

where Ω and S are the antisymmetric (rotation) and symmetric (strain) components of the velocity-gradient tensor. Positive Q identifies regions where rotation dominates strain. Each detected vortex was assigned to its nearest sensor and retained only if its core was within 32 simulation-grid cells (32 mm) of that sensor. This sensor-proximity threshold was selected heuristically by visual inspection to exclude remote structures and was not intended to represent a calibrated biological sensing distance. LilyPad uses a three-grid-cell numerical bounding region around the body. To avoid extracting vortices directly at that boundary, the baseline minimum body distance was doubled to six grid cells (6 mm). Sensitivity to thresholds of 3 and 12 mm was also evaluated.

For each sensor, a simulated pressure segment was defined as a contiguous interval of frames during which at least one retained vortex assigned to that sensor was visible. Simulated pressure values at the sensor coordinate were extracted over that interval and labelled by the generating model, fish-shaped or cylinder. The corresponding experimental sensor signal over the same time interval formed a measured pressure segment. This terminology distinguishes the 493 video samples from the multiple temporal pressure segments that could be extracted from a sample.

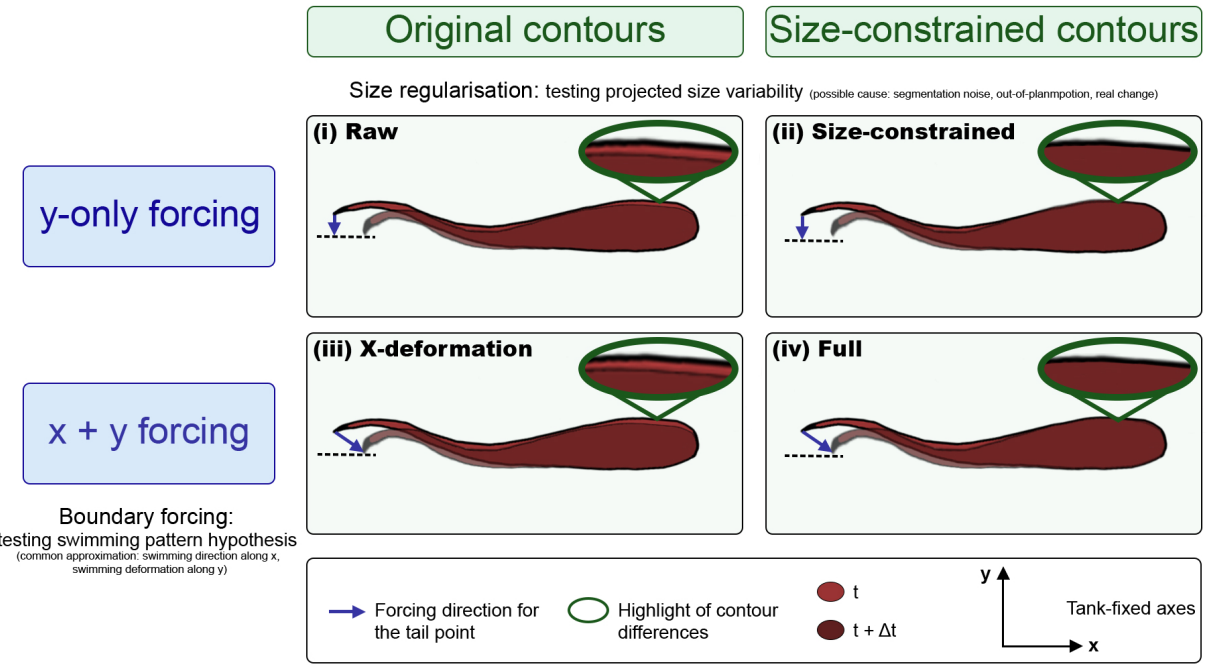


Figure 2. Illustration of the four simulation variants. Two parameters are varied: the projected contour size and the way the boundary motion is communicated to the solver. This results in four variants: the raw variant, using the original image-derived contours and y-only boundary forcing, the size-constrained variant, using a fixed profile and y-only forcing, the x-deformation variant, using the original contours and two-component forcing, and the full variant, using a fixed profile and two-component forcing. The shapes shown are conceptual illustrations of the projected contours and do not represent the actual simulation results.

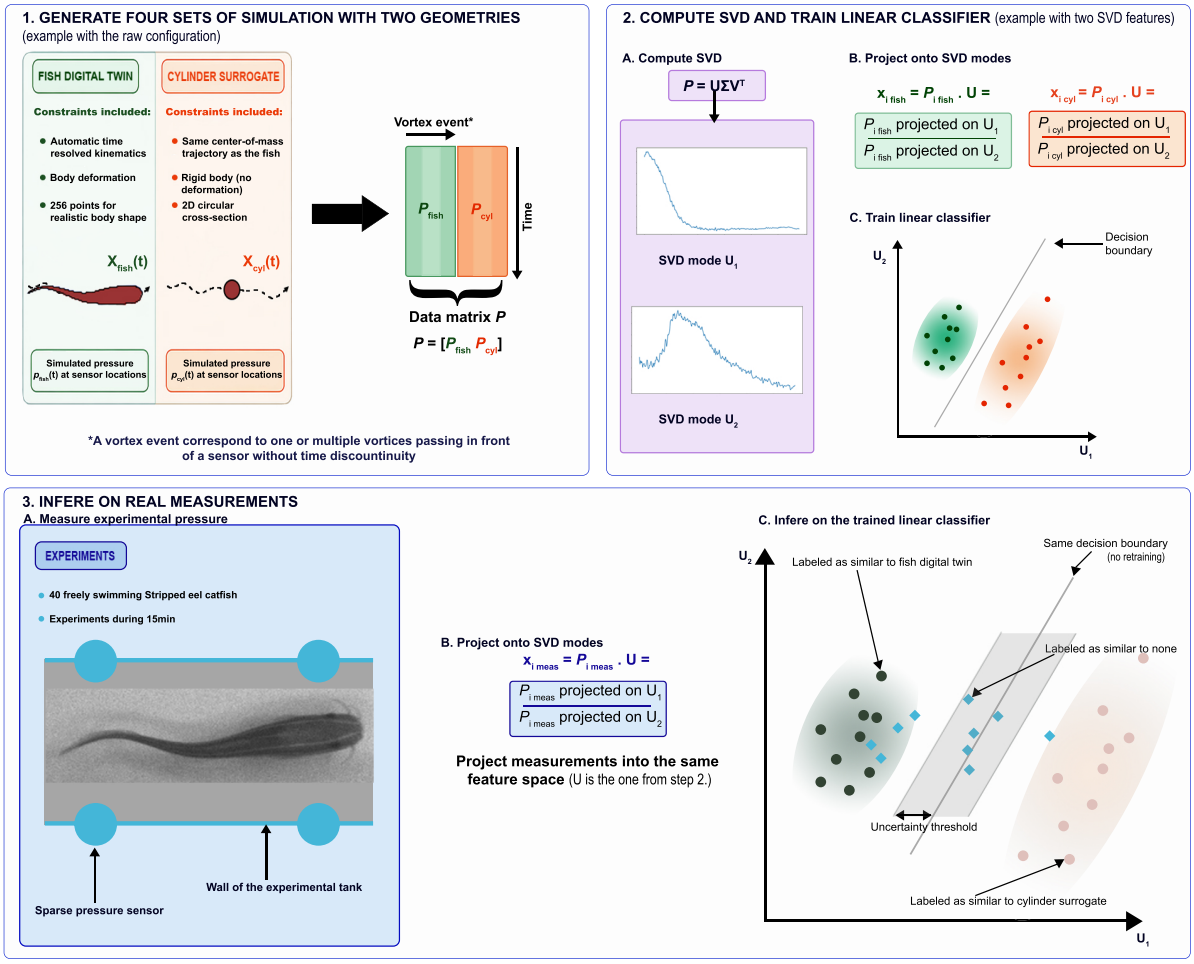


Figure 3. Schematic representation of the methodology. (1) Simulation stage: fish-shaped digital twin and cylinder surrogates are generated under different model variants (raw, size-constrained, x-deformation and full). Pressure signals are extracted from vortex tracking and assembled into a data matrix. (2) Dimensionality reduction and classification training: SVD is applied to construct a reduced basis capturing most of the dataset variance. Pressure signals are projected onto this basis to form feature vectors, which train a linear classifier to distinguish fish-like and cylinder-like simulations and define a learned similarity metric. (3) Inference: experimental pressure signals are projected onto the same feature space and evaluated with this metric.

(d) Comparative pressure-signal classification

To compare variable-duration pressure segments in a common feature space, simulated segments were mean-centred, amplitude-normalised and resampled to a common length. Singular Value Decomposition (SVD) then provided compact pressure-mode bases retaining 75%, 95% or 99% of simulated variance [25]. These three thresholds tested whether conclusions depended on the amount of pressure variation retained. A separate linear classifier was constructed for every combination of simulation variant and retained-variance level. Each classifier distinguished pressure segments generated by the fish-shaped and cylinder simulations. The deliberately simple architecture provided a task-specific, data-driven comparison score rather than a conventional distance defined a priori. Related studies have similarly learned representations or similarity and discrepancy functions from data, although using different architectures and objectives [26, 27, 28].

Measured pressure segments underwent the same preprocessing and were projected into the corresponding simulated pressure-mode space. A measured segment classified as fish-like was therefore more consistent with the fish-shaped model than with the cylinder in that feature space, not a direct reconstruction of the experimental wake. Outputs below confidence thresholds ranging from 0.50 to 0.99 were assigned to an undetermined class.

Figure 3 summarises the main classification workflow. Training settings, dataset sizes and performance are reported in supplementary sections S3–S5. Reproduction files are available at [29].

(e) Dependence among pressure segments from the same fish

Multiple measured pressure segments could originate from one video and therefore from the same fish. If within-video segments were strongly clustered, treating all segments as equivalent observations could allow a small number of fish to dominate segment-level patterns. This dependence was assessed in a measured-pressure SVD space using variance-weighted one-way intraclass correlation coefficients and same-video versus different-video distances. Details are provided in supplementary section S2.

(f) Distributional comparison of pressure-conditioned vortices

The next objective was to determine how physical vortex properties differed between the model representations associated with confident measured-pressure classifications. The analysis focused on the x-deformation variant with 95% of simulated pressure variance retained. At the 0.99 confidence threshold, fish-like measured-pressure segments were linked to vortices from the corresponding fish-shaped simulation and cylinder-like segments were linked to vortices from the corresponding cylinder simulation. Undetermined segments were excluded. These groups are therefore pressure-conditioned subsets rather than all vortices generated by either model.

Vortices were retained when their tracks contained at least 10 consecutive valid detections marked as visible within the associated pressure segment. For each track, the descriptors comprised mean and maximum area, mean and maximum equivalent-circle radius, displacement, path length, straightness, duration in detected frames, duration relative to segment length, mean and maximum speed, mean and maximum core Q-criterion magnitude, mean distance to the body and sensor, and mean core position along the tank x and y axes. Straightness was calculated as displacement divided by path length. An equivalent-circle radius was calculated from vortex area.

Descriptor distributions were first summarised across all retained vortex-track associations. A physical track intersecting more than one sensor-specific pressure segment was represented once in each relevant association. To prevent segments containing many vortices from dominating the comparison, each descriptor was then averaged within each pressure segment and the resulting segment summaries were given equal weight. Vortex count, total vortex-frame occupancy, and mean and maximum simultaneous visible-vortex count were also calculated for each segment. Finally, segment summaries were averaged within each source video and model class. Paired cylinder-minus-fish differences were calculated for videos represented in both groups. At each level, sample counts, medians and interquartile ranges were reported. At the paired-video level, where each video contributes one difference, mean differences and 95% confidence intervals from Student's t-distribution were reported as well. Further details are provided in supplementary section S8.

3. Results

(a) Dependence among segments from the same fish

Video-level dependence was quantified to determine whether repeated pressure segments allowed particular fish to dominate segment-level patterns. In a measured-pressure SVD space retaining 99.1% of variance, the variance-weighted ICC(1) was 0.058, indicating that video identity accounted for approximately 6% of retained measured-pressure variance. The median same-video distance was 7.006 and the median different-video distance was 7.443, giving a distance ratio of 0.941. Fish identity therefore contributed little to the observed pressure variation, although segments originating from the same source video should not be interpreted as independent biological replicates. Full results are reported in supplementary section S2.

(b) Classifier performance on simulated pressure segments

Before applying the learnt metric to experimental measurements, the distinguishability of pressure segments generated by the fish-shaped and cylinder simulations was tested. Test accuracies ranged from 0.55 to 0.84 across model variants and retained-variance levels. The x-deformation model achieved 0.84 at both 75% and 95% retained variance. The simulated classes were therefore separable but overlapping. These moderate accuracies support comparative use of the classifier but do not justify treating its output as direct wake reconstruction (supplementary table S4).

(c) Consistency of measured pressure signals with the models

The primary model comparison evaluated the relative consistency of measured pressure segments with the deforming fish-shaped model and the trajectory-matched cylinder, and how that consistency changed with projected-shape regularisation and full in-plane boundary motion. Figure 4 shows, for each model variant, the fractions of measured pressure segments classified as fish-like, cylinder-like or undetermined as the confidence threshold increased from 0.50 to 0.99, with 95% of simulated pressure variance retained.

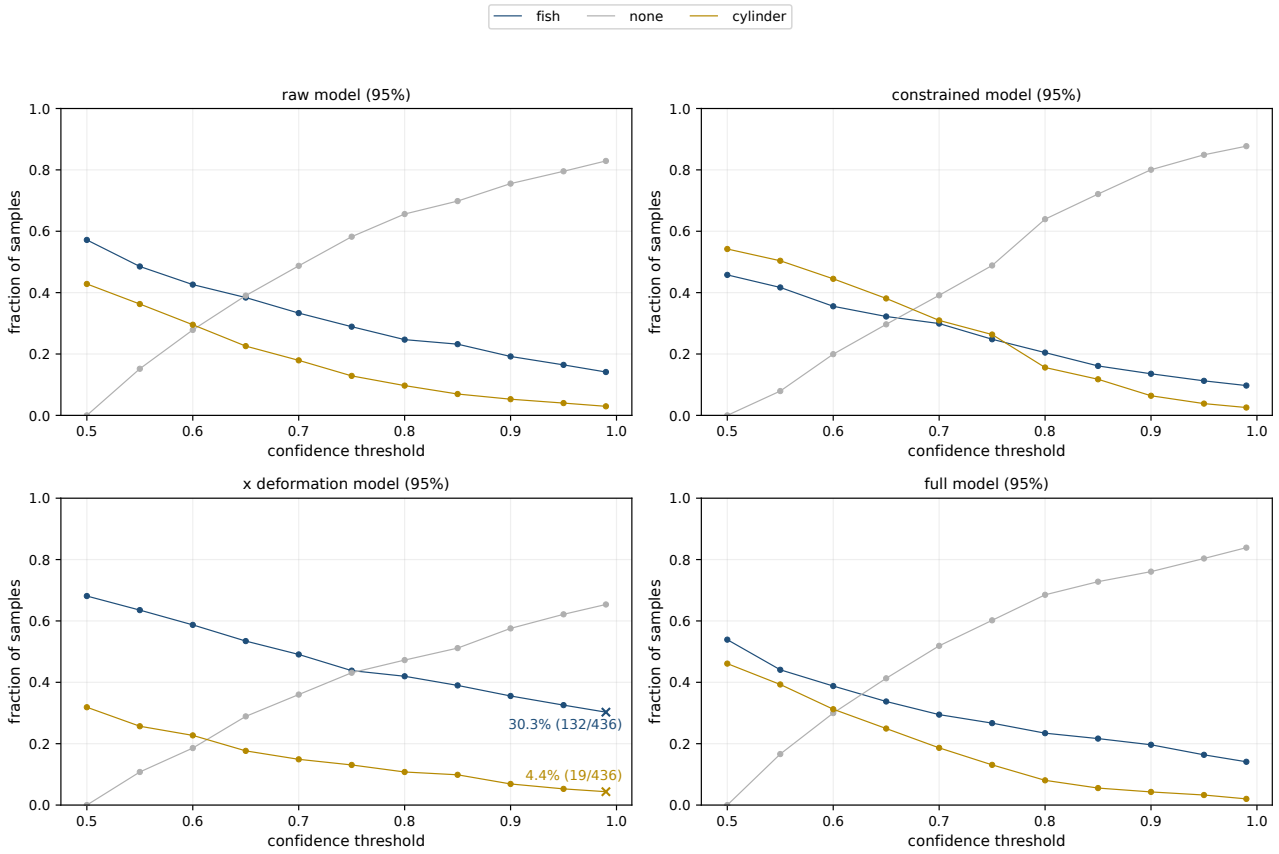


Figure 4. Classification of measured pressure segments with 95% of simulated variance retained. Panels show the raw, size-constrained, x-deformation and full variants. Blue, orange and grey curves represent fish-like, cylinder-like and undetermined fractions, respectively. Dashed lines connect the evaluated confidence thresholds and markers show individual threshold values. In the x-deformation panel, crosses and coloured labels identify the final values at a 0.99 threshold: 30.3% fish-like (132/436) and 4.4% cylinder-like (19/436).

Across most variants and confidence thresholds, measured pressure segments were classified more often as fish-like than as cylinder-like. This suggests that the hydrodynamic signature of a swimming fish depended on more than its path through the water. A cylinder following the same trajectory did not reproduce the same pressure features, whereas the deforming fish-shaped model produced signals more consistent with the experimental measurements. Although the digital twin was not a complete reconstruction of the wake, the results indicate that body profile and deformation leave detectable signatures in the pressure field.

These proportions should be interpreted alongside the simulated test accuracies of 0.55–0.84. The classifier provided a comparative measure of consistency between two overlapping model classes, particularly for variants with lower simulated test accuracy, rather than a direct probability that an experimental wake was generated by either model.

The strongest separation occurred when both x and y boundary-motion components were included without the size constraint. With the 95% retained-variance basis and a 0.99 confidence threshold, 132 of 436 measured segments were fish-like (30.3%, 95% Wilson binomial CI 26.2–34.7% [30]) and 19 were cylinder-like (4.4%, 95% Wilson binomial CI 2.8–6.7%). The remainder were undetermined. The difference between fish-like and cylinder-like fractions remained approximately constant over the evaluated confidence thresholds (Figure 4). This indicates that local in-plane deformation and motion beyond the conventional x-directed motion and y-dominated undulation contributed to the pressure signature.

In contrast, imposing the first-frame profile did not improve agreement with measured pressure. The size-constrained variant reduced separation between fish-shaped and cylinder pressure segments, and measured classifications varied more with confidence threshold. Combining the profile constraint with two-component forcing also failed to improve agreement over the unconstrained x-deformation variant.

To test whether this pattern depended on the dimensionality-reduction threshold, the analysis was repeated with bases retaining 75% and 99% of simulated variance. Both produced the same qualitative pattern (supplementary figures 3 and 4), indicating that the main comparison was not specific to the 95% basis.

(d) Fish-associated vortices were smaller, straighter and more numerous

The pressure-conditioned vortex analysis identified the physical properties associated with fish-like and cylinder-like measured-pressure segments. The pressure-conditioned subset contained 574 fish-model vortex-track associations, representing 551 distinct physical tracks from 88 measured-pressure segments and 21 source videos. It contained 65 cylinder-model associations, representing 65 distinct physical tracks from 15 segments and 10 source videos. Twenty-three fish-model tracks intersected two sensor-specific pressure segments and were therefore represented by two associations. Ten source videos were represented in both groups.

At the pooled association level, cylinder-associated vortices were larger and had higher core Q. The median track mean area was 32.1 grid cells (interquartile range 22.9–38.7) for cylinder-associated vortices and 13.8 grid cells (9.4–22.9) for fish-associated vortices. Median mean equivalent-circle radii were 3.18 grid cells (2.70–3.51) and 2.09 grid cells (1.72–2.68), respectively. Median mean core Q was 0.0331 (0.0198–0.0440) for cylinder-associated vortices and 0.00976 (0.00627–0.0165) for fish-associated vortices. Fish-associated tracks had greater median straightness at 0.825 (0.739–0.934), compared with 0.784 (0.717–0.863) for cylinder-associated tracks. Absolute displacement, path length, maximum speed and mean distance from the body showed comparatively little median separation. Absolute tracked duration was also similar between the classes, at 81.0 frames (54.0–115.0) for cylinder-associated vortices and 86.5 frames (36.0–165.0) for fish-associated vortices. Relative to the length of the pressure segment in which they were measured, however, cylinder-associated vortices persisted considerably longer, with a median duration ratio of 0.625 (0.385–0.855) compared with 0.401 (0.152–0.751). These pooled summaries give more weight to source segments that contained more retained tracks.

Giving each pressure segment equal weight preserved the differences in area, equivalent-circle radius, mean core Q, relative vortex duration and straightness. The pressure segments themselves differed markedly in length: cylinder-associated segments had a median duration of 137 frames (126.5–155.0), compared with 253.5 frames (176.0–303.5) for fish-associated segments. Because a segment was defined by the continued visibility of a retained vortex at a sensor, this difference is a property of the simulated wakes rather than a preprocessing choice, and it governs how the two duration descriptors should be read. Cylinder-associated segments contained a median of 4 retained vortex tracks (2–7), compared with 6.5 tracks (4–8) in fish-associated segments. Median total vortex-frame occupancy was also lower for cylinder-associated segments at 395 frames (216.5–525.5), compared with 678 frames (465.8–883.0). Mean simultaneous-vortex counts were similar at 2.67 (1.31–3.81) and 2.79 (2.10–3.65), respectively. Median segment-mean straightness was 0.767 (0.733–0.833) for cylinder-associated segments and 0.822 (0.765–0.871) for fish-associated segments, while path lengths were similar. Absolute segment mean tracked duration was not clearly separated at 87.6 frames (77.2–105.8) and 107.9 frames (83.1–143.2), respectively. Relative to their own segment length, however, cylinder-associated vortices occupied a substantially greater fraction, with a median duration ratio of 0.607 (0.539–0.686) compared with 0.444 (0.364–0.579). Cylinder-associated vortices therefore persisted for a similar absolute time within pressure events that were roughly half as long.

Within the ten source videos represented in both groups, paired cylinder-minus-fish differences generally agreed with the pooled and segment-informed comparisons (Figure 5). Because each video contributes a single paired difference, mean differences are reported with 95% t-based confidence intervals alongside the medians. Mean paired differences were +8.18 grid cells for mean area (95% t-based CI +2.54–+13.83, median +10.37), +0.490 grid cells for mean equivalent-circle radius (+0.15–+0.83, median +0.672), +0.0101 for mean core Q (+0.0011–+0.0190, median +0.00517), -0.0652 for straightness (-0.116– -0.014, median -0.0517) and +0.147 for duration relative to segment length (+0.020–+0.274, median +0.108). Cylinder-associated summaries also contained 2.42 fewer tracks per segment (-4.24– -0.59) and 329.5 fewer vortex frames (-534.1– -124.9), and their pressure segments were 76.9 frames shorter (-128.6– -25.2). None of these intervals included zero. Absolute tracked duration was inconclusive at this level (mean -13.2 frames, -43.1–+16.7), as it had been in the pooled and segment-informed summaries. Paired differences in displacement, path length, maximum speed, body distance, core position and simultaneous-vortex counts varied substantially among videos and their intervals included zero. These intervals are effect estimates for individual descriptors and are not adjusted for the number of descriptors examined. With ten paired videos the supporting evidence is the consistent direction across the pooled, segment-informed and video-informed summaries rather than any single interval. The agreement across analysis levels indicates that confidently cylinder-like measured-pressure segments were associated with larger vortices of greater mean core Q that occupied a greater fraction of shorter pressure events, while fish-like segments were associated with more numerous and straighter tracked vortices. These are distributional associations and do not establish that any individual vortex caused its segment-level pressure classification. Complete pooled, segment-informed and video-informed summaries are reported in supplementary tables S5–S7 and supplementary figures 1 and 2.

To determine whether the result depended on excluding vortices close to the generating body, the minimum body distance was varied from 3 to 12 grid cells for the x-deformation model (3–12 mm, Figure 6). The fish-like fraction increased when near-body structures were excluded more strongly. The baseline 6 mm filter gave 30.3% fish-like and 4.4% cylinder-like segments at the 0.99 threshold. Reducing the filter to 3 mm produced 24.8% fish-like and 10.6% cylinder-like

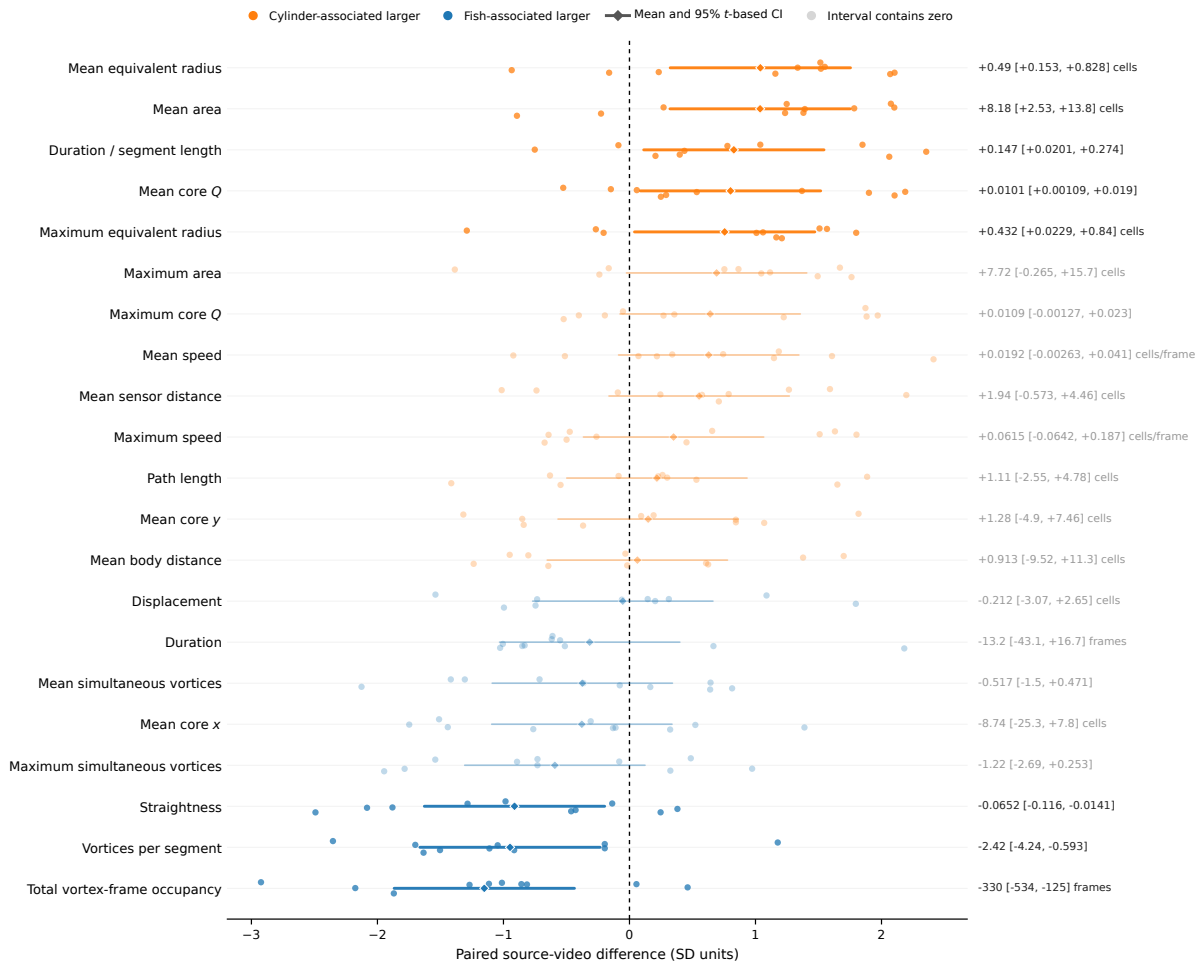


Figure 5. Paired source-video differences for pressure-conditioned vortex descriptors. Each point is the cylinder-associated source-video mean minus the fish-associated source-video mean for one of the ten videos represented in both groups. Positive values indicate larger cylinder-associated summaries and negative values indicate larger fish-associated summaries, and the dotted line marks zero difference. Because the descriptors carry different units, each difference is divided by the standard deviation of that descriptor's own ten differences so that every row shares the horizontal axis; the displacement of a row from zero therefore expresses a standardised effect size rather than a physical magnitude. Mean differences and 95% t-based confidence intervals in the original units of each descriptor are annotated on the right. Diamonds mark the mean and horizontal bars the confidence interval. Orange denotes descriptors whose cylinder-associated summary was larger and blue those whose fish-associated summary was larger; rows whose interval contains zero are drawn muted. Rows are ordered by standardised difference, from the strongest cylinder-associated contrast at the top to the strongest fish-associated contrast at the bottom. Vortices per segment, total vortex-frame occupancy and mean and maximum simultaneous vortices are segment-level descriptors; the remaining seventeen are track-level. The intervals are not adjusted for the number of descriptors examined.

classifications. Increasing it to 12 mm produced 37.9% and 7.2%, respectively. Results for all four model variants are shown in supplementary figure 5.

4. Discussion

(a) Fine-scale motion and model realism

The simulated classifiers first established that sparse pressure signals contained enough source-related information to separate the fish-shaped and cylinder models, although test accuracy varied substantially among model variants. This agrees with previous demonstrations that local flow measurements can classify canonical wakes or identify idealised obstacles [6, 7]. The present results extend that principle to pressure signals associated with measured motion from freely swimming fish. The moderate test accuracies and overlap between simulated classes remain important. Classification of measured pressure represents comparative consistency with two models rather than a reconstruction of the experimental wake or a probability that either model exactly generated the signal.

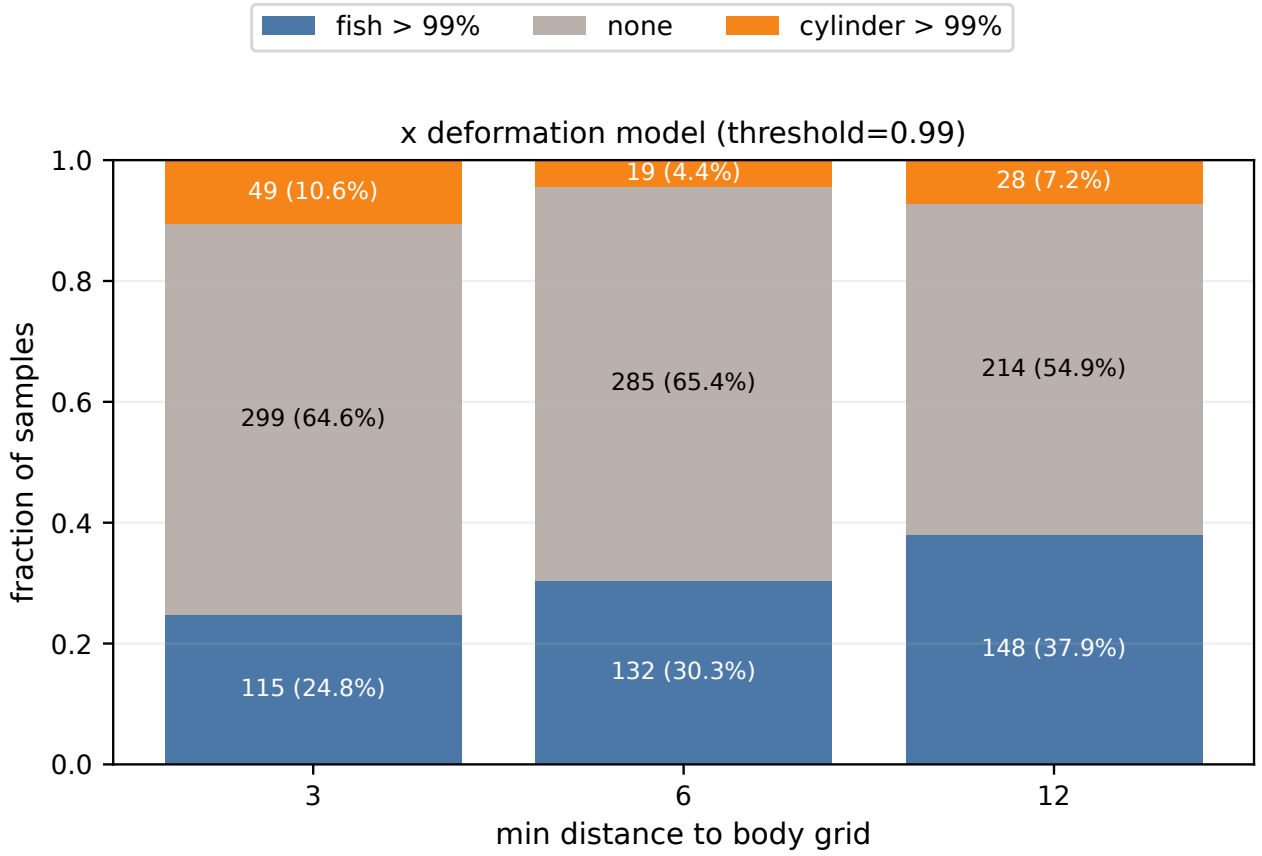


Figure 6. Effect of the minimum vortex-to-body distance on measured pressure classification for the x-deformation model at 0.99 confidence. The x-axis gives the exclusion threshold in 1 mm simulation-grid cells. Blue, grey and orange represent fish-like, undetermined and cylinder-like fractions, respectively. Increasing exclusion of near-body vortices increased the fish-like fraction.

Measured wall-pressure segments were more consistent with a deforming fish-shaped digital twin than with a cylinder following the same centre-of-mass path. Path matching alone therefore did not preserve all source-related pressure features. Controlled virtual-swimmer comparisons similarly show that body form and kinematics have distinct effects on swimming performance and smaller-scale wake structure [14]. Pressure and vorticity scaling also differ between anguilliform and carangiform swimmers, linking surrounding flow fields to body form and swimming mode [15]. The present result complements studies in which cylinders, foils, articulated swimmers and robotic fish isolate particular hydrodynamic mechanisms [10, 11, 12, 13]. Anatomically resolved simulations further show that hydrodynamic performance can depend on realistic kinematics even when trajectory and swimming conditions remain controlled [16, 17]. These models remain valuable for controlled questions, but the present comparison shows that their level of geometric and kinematic constraint matters when the target observable is pressure measured from a freely swimming fish.

The original y-only boundary forcing was a deliberate first approximation. Fish locomotion is commonly represented as progression along the swimming direction combined with a lateral body wave whose amplitude increases towards the tail [31, 19]. This approximation appeared reasonable because most trajectories followed the tank's x axis and most visible undulation occurred along y. The stronger agreement obtained with two-component forcing shows that the simplification was more consequential than expected. This supports the interpretation that apparent longitudinal redistribution of the projected body surface, local extension and compression of the projected contour, tail-tip motion towards or away from the head and other secondary in-plane movements contributed to pressure signatures reaching the sensors. Here, longitudinal and lateral refer approximately to the predominant trajectory only. The implemented components remained fixed tank-axis coordinates rather than instantaneous fish-centred axes.

(b) Size regularisation and foil-like surrogates

The poorer agreement of the size-constrained model suggests that apparent frame-to-frame contour variability was not purely segmentation noise. Because a three-dimensional fish was reconstructed from one camera, projected contour

changes may also reflect posture, roll or depth-dependent shape. Applying a fixed first-frame profile can therefore remove real projected variation together with detection error.

The constrained model can be viewed as an intermediate, foil-like representation. It retained a fish-derived profile and bending spine but imposed a fixed characteristic shape, placing it conceptually between the rigid circle and the less-constrained image-derived fish. This is an analogy rather than a third classification class. It echoes a broader limitation of flexible foils and robotic fish. Even with realistic profiles or controlled actuation [18, 19, 12, 13], regularising body size and shape may suppress posture-dependent variations that contribute to a living fish's pressure signature.

(c) Vortex persistence and spatial organisation

The pressure-conditioned vortex distributions differed most consistently in area, equivalent-circle radius, mean core Q, relative vortex duration and trajectory straightness. These differences had the same direction in the pooled, segment-informed and video-informed summaries and in the paired-video comparisons. Cylinder-associated vortices were generally larger, had greater mean core Q and occupied a greater fraction of their pressure segment, while fish-associated vortices were more numerous and followed straighter trajectories. Absolute tracked duration did not separate the two classes but the duration contrast was expressed relative to the pressure segment, which was itself markedly shorter for cylinder-associated classifications. The analysis was conditioned on confident outputs of the first pressure classifier and does not describe every vortex generated by either model.

Cylinder-associated pressure segments were concentrated in 10 source videos, compared with 21 videos for fish-associated segments. Cylinder-associated segments also contained fewer retained tracks and lower total vortex-frame occupancy, a direction that persisted in the paired-video comparison. Cylinder-associated pressure segments were also roughly half as long, while the vortices within them persisted for a similar absolute time and therefore spanned a greater fraction of the event. Combined with their larger area and greater mean core Q, these distributions raise the hypothesis that the cylinder can produce a fish-consistent pressure pattern through fewer, larger, higher-Q and relatively more persistent vortices concentrated in a brief event, while the fish-shaped model may distribute its pressure contribution among multiple smaller, lower-Q structures succeeding one another over a longer one. The analysis did not resolve the contribution of individual vortices to pressure or test pressure superposition. Because segment boundaries were set by the continued visibility of retained vortices, the detection and retention criteria influence segment length as well as the counts and durations measured within it, and tracking fragmentation could contribute further.

The distance-filter sensitivity adds context to these distributions. Reducing the exclusion distance admitted more near-body structures and increased the cylinder-like pressure fraction, whereas stronger exclusion generally increased the fish-like fraction. One possible explanation is that the cylinder reproduced some immediate disturbances associated with body passage. The change was not strictly monotonic for every class and model, so distance remains a practical spatial filter rather than a calibrated measure of vortex persistence.

This distinction is biologically relevant because a following fish encounters the wake after the generating fish has moved on. Persistent vortices can provide a longer-lived hydrodynamic signal than instantaneous near-body pressure fluctuations. This interpretation is compatible with work on schooling and vortex exploitation [5], but whether conspecifics use the specific pressure differences identified here remains an open behavioural question.

Fine-scale deformation may vary with swimming style, behavioural state, acceleration, manoeuvring, body roll or morphology, and could consequently alter downstream pressure signatures. This raises the possibility that wakes contain information about individual identity or intent. Neither possibility was tested or demonstrated here. Direct behavioural observations of followers combined with pressure measurements would be required to determine whether fish detect or use such variation.

(d) Limitations and future directions

The video reconstruction and simulations were two-dimensional approximations of a three-dimensional swimming animal. Samples where fish were not fully submerged, or where interaction with the free surface and out-of-plane motion made the two-dimensional approximation unsuitable, were excluded from the simulation analysis. Body roll, depth changes, free-surface interactions and spanwise wake structure may all affect the real pressure field. A second camera view would be required to reconstruct the three-dimensional body motion.

Three wall-mounted pressure sensors provided sparse indirect measurements. They were non-invasive but could not resolve full wake geometry. The 32 mm sensor-proximity threshold used to associate simulated vortices with sensors was a visually selected processing heuristic rather than a calibrated biological sensing distance. Future experiments should use denser pressure measurements, including sensors away from the wall, to test whether downstream fish use the detected pressure signatures in following, schooling or predator–prey contexts.

Vortex duration and trajectory were estimated from two-dimensional tracks within pressure-conditioned intervals. Tracks were required to contain at least 10 consecutive valid visible detections, but temporary tracking gaps and

fragmentation could affect duration and vortex-count summaries. Distance from the body remained a practical spatial proxy for downstream persistence. Future work should track three-dimensional vortex evolution relative to behaviourally relevant follower sensing.

Fish/video identity explained a small fraction of measured-pressure SVD variance. This limited dependence suggests that shared source-video information alone was unlikely to account for the full classification pattern. However, the classifier used a segment-level split and its performance on entirely unseen source videos was not directly evaluated. Segments originating from the same source video should therefore not be interpreted as independent biological replicates. Residual individual or recording effects may contribute to segment-level pressure signatures and should be modelled explicitly in larger datasets.

5. Conclusion

This study shows that the biological value of a fish-shaped digital twin lies not only in geometric complexity, but in preserving fine-scale body motion that can affect pressure and vortical signatures in fish wakes in ways that are detectable from sparse wall-pressure measurements. The deforming fish-shaped model was more consistent with measured pressure segments than a rigid cylinder following the same path, especially when boundary motion along both tank axes was included. Pressure-conditioned vortex distributions linked this difference to contrasts in vortex size, mean core Q, relative vortex duration and trajectory straightness. Simplified surrogates may therefore omit biologically meaningful source-related hydrodynamic information that remains detectable downstream. The same comparative approach could inform bioinspired underwater sensing and the modelling of other complex swimming bodies, as well as control strategies for underwater robotics.

The next step is to combine three-dimensional reconstruction, richer sensing and data assimilation between measurements and simulations to allow the digital twin to move beyond model comparison and towards direct inference of the hydrodynamic cues available to fish during following and predator-prey interactions.

Ethics. This study was approved by the University of Cambridge Animal Welfare and Ethical Review Body (AWERB; approval number NR2024/29).

Data Accessibility. The code and simulation files supporting this study are available at https://doi.org/10.5281/zenodo.19224932 (Lagneaux B, Porcon C, OMARINI T, Sun B, Hamilton NM, Florence R, et al. How fishes use historical hydrodynamic motion cues in search and navigation tasks: measured and simulated pressure for vortex extraction and analysis. Zenodo; 2026.). Raw experimental data (pressure signals, simulation results, and tracking files) are available from the corresponding author upon reasonable request.

Use of Artificial Intelligence (AI) and AI-assisted technologies. ChatGPT-5.5 (OpenAI) was used only for language editing, to improve readability and reduce French-influenced phrasing. The authors reviewed and edited all AI-assisted suggestions and take full responsibility for the final content.

Authors' Contributions. B.L.: conceptualization, methodology, software, formal analysis, investigation, validation, visualization, writing – original draft; C.P.: resources, methodology, investigation, writing – review and editing; M.M.: methodology, software, formal analysis, writing – review and editing; P.L.: methodology, investigation, writing – review and editing; T.O.: software, methodology, validation, writing – review and editing; B.S.: software, methodology, validation, writing – review and editing; N.M.H.: investigation, resources, data curation, writing – review and editing; F.R.: supervision, project administration, writing – review and editing; J.H.-R.: conceptualization, funding acquisition, project administration, supervision, resources, writing – review and editing; G.J.: conceptualization, methodology, funding acquisition, project administration, supervision, writing – review and editing. All authors gave final approval for publication.

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